Model Assumptions and Residual Diagnostics

Applied Linear Models I • Lecture 6 of 16

Applied Linear Models I

The assumptions are things to CHECK

Assumption	Failure	Diagnostic
Linearity	curved mean	residuals vs fitted
Homoskedasticity	variance trend	scale–location, Breusch–Pagan
Independence	autocorrelation	residual ACF, Durbin–Watson
Normality	heavy tails / skew	QQ plot

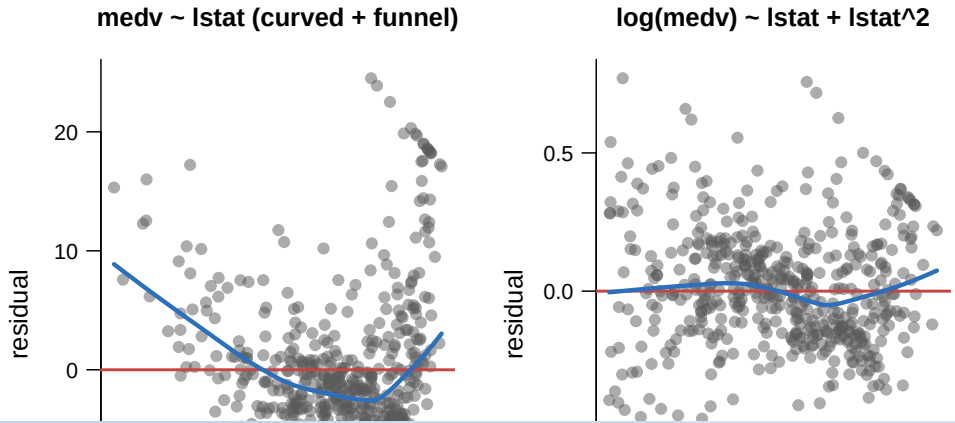
Hierarchy

Linearity & independence threaten **unbiasedness**. Heteroskedasticity & non-normality break **inference** only (OLS stays unbiased).

Residuals vs fitted

Should be a structureless horizontal band. Two failure signatures:

- a **curve** ⇒ misspecified mean (missing nonlinearity);
- a **funnel** ⇒ heteroskedasticity.

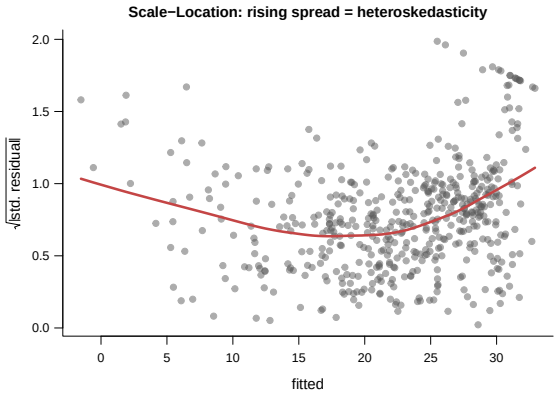


Breusch–Pagan test for heteroskedasticity

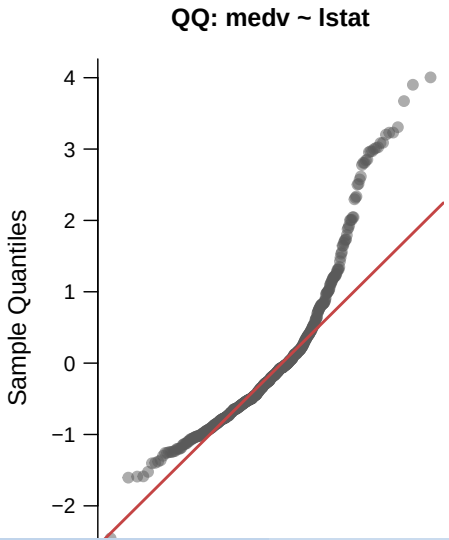
Idea: under constant variance, squared residuals are *unpredictable* from the regressors.

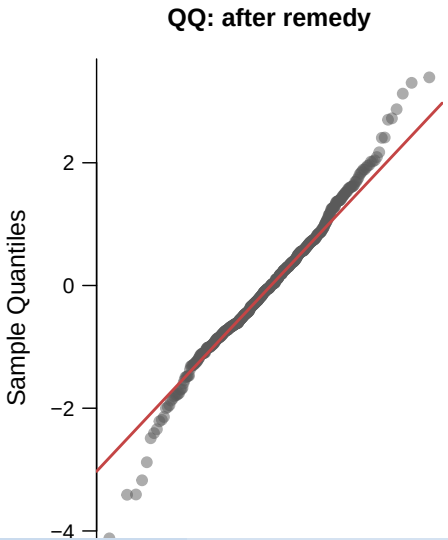
regress e_i^2 on the predictors; $nR^2_{\text{aux}} \sim \chi^2_k$ under H_0.

Boston `medv ~ lstat`: statistic ≈ 108 on 1 df, $p \approx 3 \times 10^{-25}$.



QQ plot: normality





Remedies

Robust SEs (HC3)

Keep the unbiased OLS coefficients; fix the SEs. Boston: HC3 SEs $>$ classical $\Rightarrow$ classical t-tests overstated significance.

Transformation: `log(medv) ~ lstat + lstat^2`

- curvature p: $8 \times 10^{-28} \rightarrow 0.58$ (gone);
- heteroskedasticity reduced (often a *partial* cure – re-check);
- adjusted R^2: $0.54 \rightarrow 0.68$.

Match the remedy to the violation.

Summary

- Assumptions are claims to check; linearity/independence are first-order.
- Residuals vs fitted, scale–location, Breusch–Pagan, QQ – one tool per assumption.
- HC3 SEs for heteroskedasticity; transform/respecify for curvature.
- Transformation is often a partial cure – re-run the diagnostics.

Next

Lecture 7: outliers, leverage, and influence – individual points that distort the fit.